Carbon Calculator Methodology

Repository-Derived Technical Documentation

March 26, 2026

1 Scope and Unit Conventions

This methodology documents the implemented logic in the repository for estimating monthly household greenhouse gas emissions and carbon index scores. The core reporting unit is $kgCO_2e$ per month.

- Monthly emissions are computed at activity-line level and then aggregated.
- Annual emissions are computed only for index comparison.
- Per-capita benchmark references are in tCO_2e/person/year.

2 Implemented Calculation Pipeline

2.1 Step 1: Input Capture and Normalization

Two input modes are implemented:

- **Monthly form**: user enters monthly activity values directly.
- **Weekly accessible form**: values are converted to monthly equivalents before API submission.

Weekly to monthly conversion uses:

$$\text{Monthly value} = \text{Weekly value} \times 4.345$$

Flight conversion uses:

$$\text{Monthly flight passenger-km} = \frac{\text{Flights per year} \times \text{Average flight distance (km)}}{12}$$

Accessible-form household proxies:

$$\text{Electricity}_{\text{kWh/month}} = 45 + 18 \cdot \text{household size} + 1.2 \cdot 30 \cdot \text{AC hours/day} + 25 \cdot \mathbf{1}_{\text{refrigerator}}$$

$$\text{Weekly water (L)} = 40 \cdot 7 \cdot \text{household size} + 50 \cdot \text{showers/week} + 70 \cdot \text{laundry loads/week}$$

$$\text{Water}_{\text{m}^3\text{/month}} = \frac{\text{Weekly water (L)} \times 4.345}{1000}$$

$$\text{LPG}_{\text{kg/month}} = \text{home-cooked meals/week} \times 0.08 \times 4.345$$

Diet is converted to monthly person-days as:

$$\text{Diet activity} = 30 \text{ person_day/month}$$

2.2 Step 2: Emission Factor Resolution

For each activity line (category, unit, region, date), the API resolves factors in this order:

1. **Excel custom override** from `data/emission_factors.xlsx`, when custom factor exists.
2. **Database factor** (`emission_factors` table) with region match and date validity.
3. **WORLD fallback** if region-specific factor is unavailable.
4. **Excel default row** as final fallback.

At repository state inspected on March 26, 2026, the Excel sheet contains **no active custom override rows**; effective factors therefore align with seeded defaults.

2.3 Step 3: Line Emissions and Aggregation

For each activity i:

$$E_i = A_i \times EF_i$$

where A_i is activity data in native unit and EF_i is $kgCO_2e$ per activity unit.

Total monthly footprint:

$$E_{\text{monthly}} = \sum_i E_i$$

Implementation detail:

- Each line E_i is rounded to 4 decimal places.
- Total is the sum of rounded lines, then rounded to 4 decimal places.

2.4 Step 4: Annualization and Carbon Index

Annualization:

$$E_{\text{annual, kg}} = 12 \cdot E_{\text{monthly}}$$

$$E_{\text{annual, t}} = \frac{E_{\text{annual, kg}}}{1000}$$

Index formulas:

$$\text{India Index} = \frac{E_{\text{annual, t}}}{B_{\text{India}}} \times 100$$

$$\text{World Index} = \frac{E_{\text{annual, t}}}{B_{\text{World}}} \times 100$$

Indices are rounded to 2 decimal places.

2.5 Step 5: Class-Level Benchmark Comparison (Dashboard)

Categories are grouped for class comparison:

- **Transportation**: petrol, diesel, two-wheeler, bus, metro, rail, short-haul flight.
- **Food**: diet categories + LPG.
- **Water**: water supply.

Class user annual tonnes:

$$E_{\text{class, annual, t}} = \frac{12 \cdot E_{\text{class, monthly, kg}}}{1000}$$

3 Emission Factor Table (Effective Seeded Defaults)

Activity	Category Key	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Bus	`bus_km`	WORLD	km	0.097	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diesel Car	`diesel_car_km`	WORLD	km	0.171	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diet (Meat-heavy)	`diet_meat_heavy_day`	WORLD	person_day	4.850	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Mixed)	`diet_mixed_day`	WORLD	person_day	2.750	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Plant-based)	`diet_plant_based_day`	WORLD	person_day	1.800	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Electricity	`electricity`	IN	kWh	0.757	CEA, 2025	https://www.climatiq.io/data/emission-factor/34488b8a-0e4a-47e1-a309-8 utm_source=chatgpt.com
Short-haul Flight	`flight_shorthaul`	IN	passenger_km	0.121	India GHGP, 2014	https://indiaghgp.org
Cooking Gas (LPG)	`lpg_kg`	IN	kg	2.990	Custom, 2025	https://wgbis.ces.iisc.ac.in/energy/paper/GHG_footprint/RSER_bharath2015.pdf
Metro	`metro_km`	WORLD	km	0.041	Custom, 2025	https://www.researchgate.net/publication/300425890_Energy_use_and_carbon_dioxide_emissions_assessment_in_the_lifecycle_of_passenger_rail_systems_The_case_of_the_Rio_de_Janeiro_Metro
Petrol Car	`petrol_car_km`	WORLD	km	0.171	Our World in Data, India GHGP, 2025	https://ourworldindata.org/travel-carbon-footprint

Activity	Category Key	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Rail	`rail_km`	IN	km	0.035	India GHGP, 2025	https://indiaghgp.org
Two-wheeler	`two_wheeler_km`	IN	km	0.035	India GHGP, 2025	https://www.researchgate.net/publication/343818309_India_Specific_Road_Transport_Emission_Factors
Water Supply	`water_supply_m3`	WORLD	m^3	0.344	UK Environment Agency, 2025	(URL not provided in seed data)

4 Custom Factor Derivation Methods

4.1 Diet Factors (Custom)

The repository's custom estimation note (`data/Custom_Estimation.md`) derives diet factors from food-level life cycle assessment (LCA) intensities, primarily Poore and Nemecek (2018).

General diet construction formula:

$$EF_{\text{diet}} = \sum_j \left(\text{intake}_{j,\text{kg/day}} \times EF_{j,\text{kgCO}_2\text{e/kg}}\right)$$

Raw archetype outputs documented in the repository:

$$EF_{\text{meat-heavy, raw}} = 11.9 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{mixed, raw}} = 5.85 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{plant-based, raw}} = 2.85 \text{ kgCO}_2\text{e/day}$$

After adjustment for practical consumption effects (food waste, edible portion correction, retail-versus-consumption mass, and caloric normalization), the calculator uses:

$$EF_{\text{diet, meat-heavy}} = 4.85 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, mixed}} = 2.75 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, plant-based}} = 1.80 \text{ kgCO}_2\text{e/person-day}$$

4.2 LPG Factor (Custom)

The LPG factor is derived via NCV and pollutant factors (CO_2, CH_4, N_2O), then converted to CO_2e with GWP multipliers.

General conversion:

$$EF_{\text{fuel, kgCO}_2\text{e/kg}} = \text{NCV}_{\text{TJ/kg}} \left[EF_{\text{CO}_2,\text{t/TJ}} + EF_{\text{CH}_4,\text{t/TJ}} \cdot GWP_{\text{CH}_4} + EF_{\text{N}_2\text{O},\text{t/TJ}} \cdot GWP_{\text{N}_2\text{O}}\right]$$

Repository calculation constants:

- NCV = 47.3 TJ/kt = 0.0473 TJ/kg

- $EF_{\mathrm{CO_2}} = 63.1$ t/TJ, $EF_{\mathrm{CH_4}} = 0.005$ t/TJ, $EF_{\mathrm{N_2O}} = 0.0001$ t/TJ
- $GWP_{\mathrm{CH_4}} = 28$, $GWP_{\mathrm{N_2O}} = 265$

Computed components:

$$\mathrm{CO_2} = 63.1 \times 0.0473 = 2.9836 \text{ kgCO}_2\text{/kg}$$

$$\mathrm{CH_4} \text{ as CO}_2\text{e} = (0.005 \times 0.0473) \times 28 = 0.00662 \text{ kgCO}_2\text{e/kg}$$

$$\mathrm{N_2O} \text{ as CO}_2\text{e} = (0.0001 \times 0.0473) \times 265 = 0.00125 \text{ kgCO}_2\text{e/kg}$$

$$EF_{\mathrm{LPG}} = 2.9836 + 0.00662 + 0.00125 \approx 2.99 \text{ kgCO}_2\text{e/kg}$$

4.3 Metro Factor (Custom Note)

The custom note indicates a planning range of approximately 0.03 to 0.05 kgCO$_2$e/passenger-km for electric urban rail/metro contexts, with midpoint selection rationale. The seeded operational value in this repository is:

$$EF_{\mathrm{metro}} = 0.041 \text{ kgCO}_2\text{e/km}$$

5 Benchmark Dataset Used for Indexing

Index benchmarks used by the API:

- India annual per-capita GHG benchmark: 2.9377 tCO$_2$e/person/year (2024).
- World annual per-capita GHG benchmark: 6.6691 tCO$_2$e/person/year (2024).

Additional class benchmark metrics (transport, food, water) are included for dashboard comparison and are seeded in `data/db_benchmark_stats.json`.

6 Limitations

- The model applies direct activity-factor multiplication and omits many second-order life-cycle effects.
- Several factors are scenario-based custom estimates (not universal regulatory constants), especially diet and LPG.
- Regional specificity is partial; unresolved regional gaps are intentionally filled by WORLD fallback factors.
- Benchmark years differ by metric (e.g., annual, transport, food, water), which may affect strict comparability.
- Accessible-form utility proxies are heuristic and may diverge from metered household data.

7 Tech Stack (Concise)

- **Backend**: FastAPI, SQLAlchemy ORM, Alembic migrations, Pydantic, PostgreSQL/SQLite.
- **Frontend**: Next.js (App Router), React, TypeScript, custom CSS.
- **Data Assets**: JSON seed files, editable Excel factor sheet (`data/emission_factors.xlsx`).
- **Tooling**: Pytest, npm workspaces, Docker Compose, ESLint, Prettier.

8 Repository Sources Used

- apps/api/app/services/calculation_engine.py
- apps/api/app/services/emission_factor_sheet.py
- apps/api/app/services/carbon_index.py
- apps/web/app/accessible-survey/AccessibleSurveyForm.tsx
- apps/web/app/survey/SurveyForm.tsx
- apps/web/app/dashboard/[surveyId]/DashboardClient.tsx
- data/db_emission_factors.json
- data/db_benchmark_stats.json
- data/Custom_Estimation.md
- README.md